

Final Project: 7736/39

Student: s25 | Program: NX201

2025-12-09

1. Code Analysis

System Initialization: The script uses `check_root()` to ensure sudo privileges and `install_dependencies()` to setup the environment.

Anonymity & Security: The `verify_tor_anonymity()` function confirms the IP is routed through Tor. Connection is handled via `establish_ssh_connection()`.

Scanning Logic: We use `perform_port_scan()` with Nmap (`-Pn -top-ports 200`) for fast discovery.

2. Proof of Execution (Screenshots)

```
kali@kali: ~/scripts
Session Actions Edit View Help
[2025-12-08 17:40:33] [INFO] [RECON] Running WHOIS locally
[2025-12-08 17:40:34] [SUCCESS] [RECON] WHOIS completed (local)
[2025-12-08 17:40:34] [INFO] [RECON] Running Nmap scan on 192.168.187.131
[RECON] Executing nmap port scan.... done
[2025-12-08 17:40:46] [INFO] [RECON] Running Nmap locally
[2025-12-08 17:40:51] [SUCCESS] [RECON] Nmap completed (local)
[2025-12-08 17:40:51] [INFO] [RECON] Running DNS lookup on 192.168.187.131
[RECON] Performing DNS resolution... done
[2025-12-08 17:40:56] [INFO] [RECON] Running DNS Lookup locally
[2025-12-08 17:40:56] [ERROR] [RECON] DNS Lookup failed
[2025-12-08 17:40:56] [INFO] [RECON] Running reverse DNS lookup on 192.168.187.131
[RECON] Performing reverse DNS... done
[2025-12-08 17:41:02] [INFO] [RECON] Running Reverse DNS locally
[2025-12-08 17:41:02] [ERROR] [RECON] Reverse DNS failed
[2025-12-08 17:41:02] [INFO] [RECON] Running traceroute to 192.168.187.131
[RECON] Tracing network route... done
[2025-12-08 17:41:23] [SUCCESS] [RECON] Traceroute completed (remote)
[2025-12-08 17:41:23] [INFO] [RECON] Running OS detection on 192.168.187.131
[RECON] Detecting operating system.... done
[2025-12-08 17:41:30] [INFO] [RECON] Running OS detection locally
[2025-12-08 17:41:54] [SUCCESS] [RECON] OS detection completed (local)

Report Generation
[2025-12-08 17:41:54] [INFO] [REPORT] Generating Markdown report
[2025-12-08 17:41:54] [SUCCESS] [REPORT] Markdown report created: recon_output/scan_20251208_173956/report.md
[2025-12-08 17:41:54] [INFO] [REPORT] Generating HTML report
[2025-12-08 17:41:54] [SUCCESS] [REPORT] HTML report created: recon_output/scan_20251208_173956/report.html
[2025-12-08 17:41:54] [INFO] [REPORT] Generating PDF report
[2025-12-08 17:41:56] [SUCCESS] [REPORT] PDF report created: recon_output/scan_20251208_173956/report.pdf
[2025-12-08 17:41:56] [INFO] [REPORT] Generating JSON summary
[2025-12-08 17:41:56] [SUCCESS] [REPORT] JSON summary created: recon_output/scan_20251208_173956/scan_summary.json

Scan Complete
Results saved to: recon_output/scan_20251208_173956
Execution time: 120 seconds
[2025-12-08 17:41:56] [SUCCESS] [MAIN] All operations completed successfully
[2025-12-08 17:41:56] [INFO] [REPORT] Opening HTML report in browser
[2025-12-08 17:41:56] [SUCCESS] [REPORT] HTML report opened with xdg-open
[CLEANUP] Initiating cleanup procedures
[CLEANUP] Stopping Tor service
[CLEANUP] Cleanup completed
```

```
kali@kali: ~/scripts

Session Actions Edit View Help

[2025-12-08 17:40:00] [INFO] [INIT] Dependency present: torsocks
[2025-12-08 17:40:00] [INFO] [INIT] Dependency present: jq
[2025-12-08 17:40:00] [INFO] [INIT] Dependency present: shpass
[2025-12-08 17:40:00] [INFO] [INIT] Dependency present: pandoc
[2025-12-08 17:40:00] [WARN] [INIT] Missing dependency: texlive-xetex
Installing texlive-xetex...
[2025-12-08 17:40:00] [SUCCESS] [INIT] Installed texlive-xetex
[2025-12-08 17:40:00] [WARN] [INIT] Missing dependency: texlive-fonts-recommended
Installing texlive-fonts-recommended...
[2025-12-08 17:40:01] [SUCCESS] [INIT] Installed texlive-fonts-recommended
[2025-12-08 17:40:01] [SUCCESS] [INIT] All dependencies satisfied

Tor Anonymity Layer

[2025-12-08 17:40:01] [INFO] [TOR] Starting Tor service
[TOR] Waiting for Tor to initialize... done
[2025-12-08 17:40:04] [SUCCESS] [TOR] Tor service active
[2025-12-08 17:40:05] [INFO] [TOR] Verifying Tor anonymity
[2025-12-08 17:40:05] [INFO] [TOR] Original IP: 2.54.52.42
[2025-12-08 17:40:05] [INFO] [TOR] Checking Tor IP (attempt 1)
[2025-12-08 17:40:07] [SUCCESS] [TOR] Anonymity verified - Tor IP: 88.80.26.3 (Country: Sweden)
✓ Connection is anonymous via Sweden (IP: 88.80.26.3)

SSH Connection

[2025-12-08 17:40:07] [INFO] [SSH] Establishing SSH connection to student@136.111.152.174
[2025-12-08 17:40:07] [INFO] [SSH] Connection attempt 1 of 3
[2025-12-08 17:40:10] [SUCCESS] [SSH] Connection established successfully
[2025-12-08 17:40:10] [INFO] [SSH] Retrieving jump host information
[2025-12-08 17:40:20] [INFO] [SSH] Jump host IP: 136.111.152.174
[2025-12-08 17:40:20] [INFO] [SSH] Jump host country: United States
[2025-12-08 17:40:20] [INFO] [SSH] Jump host uptime: up 51 minutes

Target Configuration

[2025-12-08 17:40:20] [INFO] [INPUT] Prompting for target address
Enter target address (IP or domain): 192.168.187.131
[2025-12-08 17:40:28] [SUCCESS] [INPUT] Target set: 192.168.187.131

Reconnaissance Operations
```

```
kali@kali: ~/scripts

Session Actions Edit View Help

LIGHTNING
REMOTE
SCANNER

[2025-12-08 17:39:59] [INFO] [MAIN] Framework execution started
[2025-12-08 17:39:59] [INFO] [INIT] Verifying root privileges
[2025-12-08 17:39:59] [SUCCESS] [INIT] Root privileges verified
[2025-12-08 17:39:59] [INFO] [INIT] Validating configuration
[2025-12-08 17:39:59] [SUCCESS] [INIT] Configuration validated
[2025-12-08 17:39:59] [INFO] [INIT] Checking system dependencies
[2025-12-08 17:40:00] [INFO] [INIT] Dependency present: nmap
[2025-12-08 17:40:00] [INFO] [INIT] Dependency present: whois
[2025-12-08 17:40:00] [INFO] [INIT] Dependency present: curl
[2025-12-08 17:40:00] [INFO] [INIT] Dependency present: tor
[2025-12-08 17:40:00] [INFO] [INIT] Dependency present: torsocks
[2025-12-08 17:40:00] [INFO] [INIT] Dependency present: jq
[2025-12-08 17:40:00] [INFO] [INIT] Dependency present: shpass
[2025-12-08 17:40:00] [INFO] [INIT] Dependency present: pandoc
[2025-12-08 17:40:00] [WARN] [INIT] Missing dependency: texlive-xetex
Installing texlive-xetex...
[2025-12-08 17:40:00] [SUCCESS] [INIT] Installed texlive-xetex
[2025-12-08 17:40:00] [WARN] [INIT] Missing dependency: texlive-fonts-recommended
Installing texlive-fonts-recommended...
[2025-12-08 17:40:01] [SUCCESS] [INIT] Installed texlive-fonts-recommended
[2025-12-08 17:40:01] [SUCCESS] [INIT] All dependencies satisfied

Tor Anonymity Layer
```

